Complex numbers

A complex number is a number written in the form

$$a + bi$$

where :

- a and b are ordinary "real" numbers,
- i is the "imaginary" number satisfying $i^2 = -1$.

Let $z = a + bi$. Then :

- a is called the real part of z and is represented by $\operatorname{Re} z$
- b is called the imaginary part of z and is represented by $\operatorname{Im} z$.

Example

- $z = 3 - 5i$ is a complex number. We have $\operatorname{Re} z = 3$ and $\operatorname{Im} z = -5$.
- $z = -1 + i$ is a complex number. We have $\operatorname{Re} z = -1$ and $\operatorname{Im} z = 1$.
- $-i$ is a complex number. We have $\operatorname{Re} z = 0$ and $\operatorname{Im} z = -1$.

Addition of complex numbers

How do we add complex numbers?

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Example

- Suppose we want to add $3 - i$ and $1 + 2i$.

$$(3 - i) + (1 + 2i) = (3 + 1) + (-1 + 2)i = 4 + i$$

- Suppose we want to add $-1 + i$ and $2 + 3i$.

$$(-1 + i) + (2 + 3i) = (-1 + 2) + (1 + 3)i = 1 + 4i$$

- Suppose we want to add $i - 2$ and $1 - 2i$.

We first note that $i - 2 = -2 + i$. Then :

$$(-2 + i) + (1 - 2i) = (-2 + 1) + (1 - 2)i = -1 - i$$

- We can subtract complex numbers similarly :

Subtraction of complex numbers

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

Example.
$(3 - 5i) - (2 - 4i) = (3 - 2) + (-5 - (-4))i = 1 - i$

Complex conjugation

For a complex number $z = a + bi$, the complex conjugate of z is defined as

$$\overline{z} = a - bi$$

Example.
$\overline{3 - 2i} = 3 + 2i$

$\overline{\frac{1}{2} + \sqrt{3}i} = \frac{1}{2} - \sqrt{3}i$

Multiplication of complex numbers

How do we multiply complex numbers?

$$(a+bi)(c+di) = ac + adi + bic + bidi = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$$

Example

$$(2-3i)(-1+2i) = 2(-1) + 2(2)i + (-3)(-1)i + (-3)(2)i^2 = -2 + 4i + 3i + 6 = 4 + 7i$$

$$(i-1)(3-i) = 3i + i(-i) + (-1)(3) + (-1)(-i) = 3i - i^2 - 3 + i = -2 + 4i$$

$$i^4 = i \cdot i \cdot i \cdot i = (-1) \cdot i \cdot i = (-1)(-1) = 1$$

Absolute value

Consider a complex number $z = a + bi$. The absolute value (or modulus or norm) of z is defined by

$$|z| = \sqrt{a^2 + b^2}$$

Note that $z\overline{z} = (a + bi)(a - bi) = a^2 + b^2$. Therefore we have

$$|z| = \sqrt{z\overline{z}}$$

- The absolute value of a complex number is always a nonnegative real number.

Example.

$|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

$|-1 - 2i| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$

Note that :

$$|ai| = \sqrt{a^2} = |a|$$

For instance $|-5i| = \sqrt{(-5)^2} = \sqrt{25} = 5.$

Basic properties of complex numbers

Let z, z_1, and z_2 be complex numbers. Then :

1. $\overline{z} = z$ if and only if z is a real number, i.e., $\operatorname{Im} z = 0$.
2. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
3. $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$
4. $z\overline{z} = |z|^2 \geqslant 0$.
5. $|z_1 z_2| = |z_1||z_2|$
6. Triangle Inequality : $|z_1 + z_2| \leqslant |z_1| + |z_2|$

Division of complex numbers

Example

Simplify $\dfrac{1+3i}{-1+5i}$.

Solution.

$$\frac{1+3i}{-1+5i} = \frac{(1+3i)(-1-5i)}{(-1+5i)(-1-5i)} = \frac{-1-5i-3i-15i^2}{1+5i-5i-25i^2} = \frac{14-8i}{1-(-25)} = \frac{14-8i}{26}$$

$$= \frac{1}{26}(14-8i) = \frac{14}{26} - \frac{8}{26}i$$

Division of complex numbers

In general, if $z = a + bi$ and $w = c + di$ are complex numbers and $w \neq 0$, then $\frac{z}{w}$ is obtained as follows :

$$\frac{z}{w} = \frac{z\overline{w}}{w\overline{w}} = \frac{1}{|w|^2}(z\overline{w})$$

Example

$$\frac{3-i}{-1+i} = \frac{(3-i)(-1-i)}{(-1+i)(-1-i)} = \frac{-3-3i+(-i)(-1)+(-i)^2}{1+i-i-i^2}$$
$$= \frac{-3-3i+i-1}{1+i-i+1} = \frac{-4-2i}{2} = -2-i$$

In particular, using the above method one can see that if $z = a + bi$ is a complex number then

$$\frac{1}{z} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$$

Example. Simplify $\frac{1}{-2+i}$.

Solution. $\frac{-2}{((-2)^2+1^2)} - \frac{1}{(-2)^2+1^2}i = -\frac{2}{5} - \frac{1}{5}i.$

You can check that

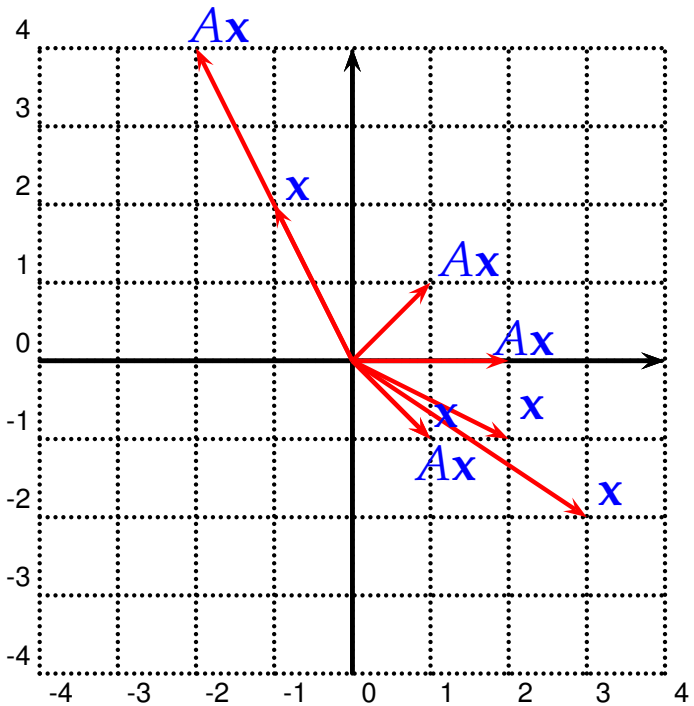
$$(-2+i)(-\frac{2}{5} - \frac{1}{5}i) = 1$$

http://www.youtube.com/watch?v=aP4fWMLofvo

Eigenvalues and Eigenvectors

Problem. Consider the matrix $A = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$. What does multiplication by A do to vectors in $\mathbb{R}^2$?

$$\mathbf{x} \mapsto A\mathbf{x}$$



$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

- We are interested in directions which remain invariant by the matrix.
- This means we want to find vectors $\mathbf{x}$ in $\mathbb{R}^n$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ.

Let A be an $n \times n$ matrix.

If a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A corresponding to λ.

Example

$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$ corresponding to $\lambda = 2$:

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Remark

The eigenvector must be nonzero, but the eigenvalue can be zero or nonzero.

Let A be an $n \times n$ matrix.

If a nonzero vector $\mathbf{x}$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A corresponding to λ.

Example

- Is $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ an eigevector of $\begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}$?

No! Because $\begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -5 \\ -6 \end{bmatrix}$ and there is no λ such that $\begin{bmatrix} -5 \\ -6 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Example

- The vector $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 1 & 1 & 0 \\ 3 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}$ corresponding to $\lambda = 0$

because $\begin{bmatrix} 1 & 1 & 0 \\ 3 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

Let A be an $n \times n$ matrix.

If a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A corresponding to λ.

Example

The vectors $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}-2\\-2\end{bmatrix}, \begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix}$ are eigenvectors of $\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}$ corresponding to $\lambda = 3$:

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\3\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}-2\\-2\end{bmatrix} = \begin{bmatrix}-6\\-6\end{bmatrix} = 3\begin{bmatrix}-2\\-2\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix} = \begin{bmatrix}3\sqrt{3}\\3\sqrt{3}\end{bmatrix} = 3\begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix}$$

If $\mathbf{x}$ is an eigenvector of A corresponding to λ, then for every scalar $c \neq 0$, the vector $c\mathbf{x}$ is also an eigenvector of A corresponding to λ.

Reason : $A(c\mathbf{x}) = c(A\mathbf{x}) = c\lambda\mathbf{x} = \lambda(c\mathbf{x})$.

Examples

Example

Is -2 an eigenvalue of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$?

Solution.

$A\mathbf{x} = (-2)\mathbf{x} \implies A\mathbf{x} - (-2)\mathbf{x} = \mathbf{0} \implies A\mathbf{x} + 2\mathbf{x} = \mathbf{0} \implies (A + 2\,\mathrm{I})\mathbf{x} = \mathbf{0}$

where $\mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

$$A + 2\mathrm{I} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$$

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow$ we should solve the homogeneous linear system $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{cc|c} 3 & 3 & 0 \\ 3 & 3 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 3 & 3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$\implies \{x_1 + x_2 = 0$ $\boxed{x_1 : \text{basic}, x_2 : \text{free.}}$

$x_1 + x_2 = 0 \implies x_1 = -x_2 \Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Examples

Example

Is -2 an eigenvalue of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$?

Recall: $A\mathbf{x} = (-2)\mathbf{x} \implies (A + 2\mathrm{I})\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

It follows that -2 is an eigenvalue of A.

Any vector of the form $c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, with $c \neq 0$, is an eigenvector of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ corresponding to $\lambda = -2$.

One can check that $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = (-2) \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Examples

Example

Is 4 an eigenvalue of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$?

Solution.

$$A\mathbf{x} = 4\mathbf{x} \quad \Longrightarrow \quad A\mathbf{x} - 4\mathbf{x} = \mathbf{0} \quad \Longrightarrow \quad (A - 4\,\mathrm{I})\mathbf{x} = \mathbf{0} \quad \text{where } \mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$A - 4\,\mathrm{I} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$$

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow$ we should solve the homogeneous system $\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{cc|c} -3 & 3 & 0 \\ 3 & -3 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 3 & -3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$\Longrightarrow \quad \{x_1 - x_2 = 0$ $\boxed{x_1 : \text{basic}, x_2 : \text{free.}}$

$$x_1 - x_2 = 0 \quad \Longrightarrow \quad x_1 = x_2 \quad \Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Examples

Example

Is 4 an eigenvalue of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$?

Recall: $A\mathbf{x} = 4\mathbf{x} \implies (A - 4\,\mathrm{I})\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

It follows that 4 is an eigenvlue of A.

Any vector of the form $c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, with $c \neq 0$, is an eigenvector of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ corresponding to $\lambda = 4$.

One can check that $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Examples

Example

Is 2 an eigenvalue of $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$?

$$A\mathbf{x} = 2\mathbf{x} \quad \Longrightarrow \quad A\mathbf{x} - 2\mathbf{x} = \mathbf{0} \quad \Rightarrow \quad (A - 2\mathrm{I})\mathbf{x} = \mathbf{0}.$$

$$A - 2\mathrm{I} = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 2 & 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -1 & 3 & 0 \\ 2 & 3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{cc|c} -1 & 3 & 0 \\ 0 & 9 & 0 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{9}R_2} \left[\begin{array}{cc|c} -1 & 3 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 3R_2} \left[\begin{array}{cc|c} -1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \to -R_1} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

There are no free variables.

Therefore the homogeneous system has a unique solution, which is the trivial solution $\mathbf{x} = \mathbf{0}$.

It follows that 2 is NOT an eigenvalue of $\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$

Theorem

Let A be an $n \times n$ matrix.

- A scalar λ is an eigenvalue of A if and only if the homogeneous linear system

$$(A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$

has nontrivial solutions.

- Here "I" denotes the $n \times n$ identity matrix.

Eigenspaces

Let A be an $n \times n$ matrix.

If λ is an eigenvalue of A, the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ satisfying

$$A\mathbf{x} = \lambda\mathbf{x}$$

forms a subspace of $\mathbb{R}^n$. It is called the eigenspace of A corresponding to λ.

- Indeed the eigenspace corresponding to λ is identical to $\mathrm{Nul}(A - \lambda \mathrm{I})$.

$$A\mathbf{x} = \lambda\mathbf{x} \Leftrightarrow (A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$

Example

Is -1 an eigenvalue of $A = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -1 \\ 2 & 1 & 0 \end{bmatrix}$? If so, then determine a basis for the eigenspace corresponding to -1.

Solution.

$A\mathbf{x} = -\mathbf{x} \Rightarrow A\mathbf{x} + \mathbf{x} = \mathbf{0} \Rightarrow (A + \mathrm{I})\mathbf{x} = \mathbf{0}$.

$$A + \mathrm{I} = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -1 \\ 2 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ -2 & -1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ -2 & -1 & -1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The system has nontrivial solutions $\Longrightarrow \lambda = -1$ is an eigenvalue of A.

$$x_1 + \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 = 0 \Rightarrow x_1 = -\tfrac{1}{2}x_2 - \tfrac{1}{2}x_3 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_2 - \frac{1}{2}x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

$\left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = -1$.

Example

Is 3 an eigenvalue of $A = \begin{bmatrix} 4 & -1 & 0 \\ 1 & 4 & -1 \\ 2 & -1 & \frac{5}{2} \end{bmatrix}$? If so, then determine a basis for the eigenspace corresponding to 3.

Solution.

$A\mathbf{x} = 3\mathbf{x} \Rightarrow A\mathbf{x} - 3\mathbf{x} = \mathbf{0} \Rightarrow (A - 3\mathrm{I})\mathbf{x} = \mathbf{0}$.

$$A - 3\mathrm{I} = \begin{bmatrix} 4 & -1 & 0 \\ 1 & 4 & -1 \\ 2 & -1 & \frac{5}{2} \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & -1 \\ 2 & -1 & -\frac{1}{2} \end{bmatrix} \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 2 & -1 & -\frac{1}{2} & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_2]{\substack{R_2 \to \frac{1}{2}R_2 \\ R_1 \to R_1 + R_2}} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The system has nontrivial solutions $\Longrightarrow \lambda = 3$ is an eigenvalue of A.

$$\begin{cases} x_1 = \frac{1}{2}x_3 \\ x_2 = \frac{1}{2}x_3 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_3 \\ \frac{1}{2}x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}$$

$\left\{ \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = 3$.

Recall : Triangular matrices

- An $n \times n$ matrix is called upper triangular if every entry below the diagonal is zero.

$$\begin{bmatrix} -1 & 3 & -1 & 3 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

- An $n \times n$ matrix is called lower triangular if every entry above the diagonal is zero.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 & 0 \\ 3 & 0 & -4 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & -3 & 4 & 5 & -1 \end{bmatrix}$$

- A matrix is called triangular if it is either upper triangular or lower triangular.

Eigenvalues of a triangular matrix

Theorem

The eigenvalues of a triangular matrix are the diagonal entries of the matrix.

Example. The eigenvalues of $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & -3 & 4 & 5 & -1 \end{bmatrix}$ are -1, 0, 1, 2.

We disregard repetitions.

Eigenvalues of powers of a matrix

Theorem

Let A be a square matrix whose eigenvalues are $\lambda_1, ..., \lambda_k$. Then the eigenvalues of A^r are $\lambda_1^r, ..., \lambda_k^r$.

Key idea : If $A\mathbf{x} = \lambda\mathbf{x}$ then :

$$A^2\mathbf{x} = A(A\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda(\lambda\mathbf{x}) = \lambda^2\mathbf{x}$$

Similarly $A^3\mathbf{x} = A(A(A\mathbf{x})) = \lambda^3\mathbf{x}$, etc.

Example If $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & 5 & 4 & 5 & -1 \end{bmatrix}$, what are the eigenvalues of A^3 and A^4?

Solution.

- Since A is triangular, the eigenvalues of A are its diagonal entries, i.e., -1, 0, 1, 2.
- The eigenvalues of A^3 are $(-1)^3, 0^3, 1^3, 2^3$; i.e., -1, 0, 1, 8.
- The eigenvalues of A^4 are $(-1)^4, 0^4, 1^4, 2^4$; i.e., 0, 1, 16.

Example

Let $A = \begin{bmatrix} i & -1+i & 1+2i & 4i-3 \\ 0 & -i & 4+i & -5 \\ 0 & 0 & -1-i & 2 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

- What are the eigenvalues of A?

Solution. The eigenvalues of A are $i, -i, -1-i, -1$.

- What are the eigenvalues of A^2?

Solution. The eigenvalues of A^2 are $i^2 = -1$, $(-i)^2 = -1$, $(-1-i)^2 = 2i$, $(-1)^2 = 1$.

- What are the eigenvalues of A^{-1}?

Solution. The eigenvalues of A^{-1} are $\frac{1}{i} = -i$, $\frac{1}{-i} = i$, $\frac{1}{-1-i} = -\frac{1}{2} + \frac{1}{2}i$, and $\frac{1}{-1} = -1$.

Summary

Let A be an $n \times n$ matrix.

If a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A corresponding to λ.

Theorem

Let A be an $n \times n$ matrix.

- A scalar λ is an eigenvalue of A if and only if the homogeneous linear system

$$(A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$

has nontrivial solutions.

- Here "I" denotes the $n \times n$ identity matrix.

Summary

Eigenspaces

Let A be an $n \times n$ matrix.

If λ is an eigenvalue of A, the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ satisfying

$$A\mathbf{x} = \lambda\mathbf{x}$$

forms a subspace of $\mathbb{R}^n$. It is called the eigenspace of A corresponding to λ.

- Indeed the eigenspace corresponding to λ is identical to $\mathrm{Nul}(A - \lambda \mathrm{I})$.

$$A\mathbf{x} = \lambda\mathbf{x} \Leftrightarrow (A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$